Richco International Inc
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TRAFFIC CONTROL AND WORK ZONE SAFETY PROGRAM

Overview

The number of persons killed in the United States in motor vehicle crashes in work zones continue to rise. There are now in excess of 1,000 deaths annually. In addition, more than 40,000 people are injured each year as a result of vehicle crashes in a work zone. It is for this reason that we have established this Traffic Control and Work Zone Safety Program. Although the nature of our operations does not usually require freeway lane closures, we often find that unloading material, installing conduit in excavation, overhead lighting installation and other work activities cause employees to be adjacent to or in actual vehicular traffic. Further, our work may also expose pedestrians or bicycle traffic to hazard, and this program will address that issue as well. Obviously, we are concerned that our materials, tools, equipment, etc. could cause damage to vehicles or cause injury to the general public when our work encroaches into accessible areas. But we are most concerned about the health and safety of our employees who may work in these hazardous conditions.

Program Objectives

- Our first and most important objective is to ensure a high level of safety for our employees, other workers, and the public.
- Our second objective is to minimize adverse traffic impacts associated with construction in terms of delays and congestion, noise, and other environmental impacts, to ensure a high level of satisfaction by our ultimate customer – the public.
- Our third objective is the identification of specific risk factors associated with individual projects that present traffic hazard conditions and the development of countermeasures to mitigate those hazards to minimize the risk.

Program Goals

- No employees or other workers will be seriously injured as a result of traffic accidents in our work zones, regardless of the cause of the accident.
- No members of the public traveling through our work sites will be injured as a result of traffic accidents caused by or made more serious by our work or traffic control operations.
- Traffic delays and congestion associated with our work activities will be held to the minimum level that can be achieved, consistent with the traffic volumes through the project, the characteristics of the roadway where the work occurs, and the nature of the work operations that must be completed.
- Interference with businesses and other activities adjacent to the project will be held to a

Minimum, consistent with traffic volumes, roadway characteristics and work operations.

- Traffic controls will be carried out in a manner that ensures that emergency services in and through the project can be delivered in a timely manner, and transit operations will not be disrupted.
- Traffic controls will be conducted in a manner that provides us with adequate access to the work area to complete the planned work on schedule, in compliance with quality requirements of the contract. Efficient delivery of materials and equipment to work sites will be available.
- All traffic control operations will be in full compliance with the terms and requirements of the construction contract and/or in keeping with current States and Federal traffic control standards. Full understanding of the Manual on Uniform Traffic Control Devices (MUTCD) may be necessary to fulfill these requirements. Temporary Traffic Control Part 6 refers to the traffic control procedures and methods to be followed, and additional state addendums may apply.
- All traffic control devices, safety features, and work zone operations shall be maintained in good condition at all times, with full documentation of all work zones operations recorded according to the requirements of this Work Zone Program.

Regulatory and Contractual Requirements

1. Each project will have a particular traffic control demands and needs; will determine issues related to work within a traffic control zone. Delivery of materials in public roadway locations or adjacent to roadway locations or adjacent to roadways will be determined and scheduling and planning will be accomplished to minimize hazard.
2. Any foreman or superintendent with direct work zone responsibilities will have access to the Federal MUTCD and appropriate State supplementary requirements before the start of the project.
3. Typically, the General Contractor is responsible for traffic control on the construction site and coordination with the temporary traffic control plan of the General Contractor is necessary. Additional traffic control planning may be required as determined by the scope of work.
4. OSHA regulations indicate that signs, barricades, and flagger signaling activities comply with the MUTCD regulations. Violations of the MUTCD requirements in these areas could be fully citable by OSHA.

5. Vehicle and traffic laws of the particular state and municipality must be followed, and permits may be required for lane or sidewalk restriction or closures. All such issues should be addressed in pre-planning meetings and in conjunction with the General Contractor.
6. Department of Transportation regulations in each State may require additional measures taken and this must be coordinated at time of job start up.

Worker Visibility

1. A critical element of work zone safety is high visibility of workers when adjacent to traffic.
2. All workers must wear high visibility apparel when in or adjacent to traffic. Class 1, 2 or 3 garments will be used based upon specific exposure needs.
3. Working during dawn or dusk conditions will require use of fluorescent colored high visibility apparel (class 2 or 3 based upon exposure.
4. Yellow green apparel should be considered when a significant amount of orange-colored signs, drums or vehicles are in the work zone.
5. Employees will be issued appropriate high visibility vests based upon exposure. This clothing must be kept in good condition and laundered periodically to assure adequate visibility. Clothing/vests should be replaced as needed to assure color discrimination by motorist.

Worker Traffic Zone Training

1. All employees exposed to traffic will be t rained in how to work safely near traffic.
2. Any temporary traffic control measures implemented will only be complete by employees who have been fully trained and are cognizant of MUTCD requirements.
3. Any specific work procedures will be established by the Superintendent, General Foreman or Foreman when work activities present traffic hazards.
4. Traffic control plans will be clearly communicated to all affected personnel.
5. When personnel are responsible for traffic control device maintenance, those employees will be fully trained.

Area Planning

1. Routes for work activity and traffic must be clearly established and marked properly. A control plan will be established to assure all parties are aware of encroachment and closures.
2. Safe access and egress of workers and vehicles will be determined.
3. Backing of vehicles should involve back up alarms and spotters whenever pedestrians or workers may be present.
4. Overhead and underground utilities should be located and marked to prevent contact by equipment and workers.

Traffic Speed Control

1. Compliance with posted speed limits is vital to protect workers and the traveling public. Reduction of initial posted speed limits may be required based upon job demands and work tasks.
2. Appropriate speed limits should be established and application for reductions should be coordinated with the General Contractor or whoever is responsible for the traffic control plan.
3. Assure that the determined regulatory speed is properly and clearly posted immediately preceding the work zone.
4. Flaggers may be needed to monitor and slow traffic as necessary.
5. Local law enforcement may be needed to enforce speed limits. Coordinate with the General Contractor to determine if law enforcement presence will be needed at the site.

Separation of Traffic and Work Activities

1. Effective separation of traffic from work activity is one of the most critical aspects of work zone safety.
2. Use of temporary traffic barriers, shadow vehicles or other devices must be determined and implemented according to traffic control plan.
3. Factors to consider when determining what separation is needed are as follows:

- Traffic speed
- Traffic volume
- Distance between workers and vehicles
- Duration of work

- Type of work operations

- Physical hazards within the work zone
- Alignment of traffic lanes through the work zone

4. Coordination with General Contractor is critical to determine work areas that require separation. Other contractors work activity may come into play when determining the work zone.
5. Adequate lateral buffer space should be provided to the greatest extent possible by shifting traffic slightly away from work operation as much as available space permits.
6. Where workers must work very close to traffic, a trained flagger or spotter should be used to protect workers from oncoming traffic.
7. For short duration work activity like pothole patching, taking measurements, marking, etc. the operation should be planned using a flagger or at least a spotter to make sure workers are not exposed to traffic in open lanes without protection.
8. Workers must not cross high speed travel lanes on foot unless the foremen analyze the situation and develops procedures to accommodate movement safely.
9. Warning sign should be posted at locations where work vehicles regularly enter and exit traffic.

Technical Requirements for Work Zone Temporary Traffic Control

1. Basic requirements are detailed in the MUTCD. This manual should be consulted when setting up a control plan.
2. Flagging requirements, if applicable, mandate worker training and this should be established upon job start up.
3. All manuals of traffic control will be retained on the job for ongoing review.
4. Copies of any established traffic control plans will be maintained and reviewed with effected employees.
5. Required signs will be determined and an adequate supply ordered to come onto the job site in advance of traffic control needs. Reasonable damage should be considered and additional quantities should be available.
6. All necessary sign supports will be ordered in quantity to allow for damage and vandalism.
7. Channelizing devices (cones, drums, and barricades) must be ordered as required. Specification of 28” and 36” cones should be established to make certain that required cones are used based upon speed requirements.

Type III barricades when required should be in adequate quantity to assure safe and complete closures.

8. Warning lights and batteries will be maintained on site when night closure or overnight securing is needed.
9. Portable concrete barrier (k-rail) will be used when required by contract or long-term protection of workers in the work zone is needed. Appropriate warning and signage will be required.
10. All barriers must be in good condition to present displacement when impacted.
11. Vehicles will be installed with warning beacons, flashers and markings as required in accordance with MUTC requirements.
12. Reflective markings and beacons must be kept in good condition and cleaned often to assure maximum visibility.

Work Zone Layout

- Many situations follow "typical"" MUTCD requirements and details should be followed specifically. The type of Temporary Traffic Control Guide layout should be clearly identified in the traffic control plan and appropriate page, or diagram reference should be made to MUTCD.
- Contract specifications or permit documents will typically include drawings or diagrams showing setups to be used for lane closures, flagger stations, shoulder closures and other typical set ups. This specification must be followed and referenced in foreman daily logs.
- Significant deviations from contract or permit layouts should be reviewed with General Contractor and signed off before implementation.
- Almost all layouts will require some minor modifications based upon actual site conditions. The foreman in charge will make the minor adjustments required, but it is expected that changes err on side of additional warnings and be in the best judgment of the foreman. Any changes should be documented on the daily log. These adjustments would include:
    1. Minor shifts in sign location (10 percent or less) to avoid driveways, intersections, other devices and to improve visibility.
    2. Minor shifts in start point and length of tapers (10 percent on length, 100 ft on start point to avoid intersections and other conflicts and to improve sight distance.
    3. Add supplemental signs on the left side of the road to improve visibility.
    4. Reduce spacing of channelizing devices at potential problem areas

5. Other minor changes to address obvious concerns.

- Any non-typical situation should be reviewed with a traffic control qualified person or engineer and the layout document should be reviewed with a traffic control qualified person or engineer and the layout document should be fully documented. Written plan should be maintained on the site for ongoing crew review for adherence to requirements.
- Additional conditions may require further devices, including “pavement ends” or “bump” signs. These should be provided and documented in the plan.

Installing and Removing Traffic Control Set Ups

1. Periods during placement and removal of traffic control devices often present the highest risk of accident for workers and public because devices are not completely in place and workers and equipment are most exposed to traffic.
2. Proper planning and careful execution of set ups and removals is needed, adhering to established procedures for protection of workers.
3. Common sense and use of spotter or other means of identifying potential hazards are needed at all times when placement and removal.
4. The following steps are required when setup/removal operations:

- All fixed signs must be placed first, with covers attached so signs can be covered and uncovered quickly at the appropriate time. Covers will be placed at the end of the operation.
- Portable signs will be deployed in the necessary location along the route.
- Channelizing devices will be set into place in the proper pattern. Note carefully the number of cones and angles of tapers and follow plan.
- If setting devices on foot make certain that a shadow vehicle is in place or worker carefully observes oncoming traffic.
- Workers must not cross open lanes of high-speed traffic to place devices.
- Arrow boards will be place in the proper sequence as cones are placed. Normally, these are towed into place and should be activated when other devices are set.
- Set up operations normally should start at the approach end and proceed downstream in the direction of traffic flow unless conditions determined by Superintendent or foreman dictate other measures and more safe.
- Removal will normally proceed in the opposite direction of set up.
- The foreman will be responsible for driving through all set ups after completion to make sure it meets all requirements.

- The foreman is also responsible for driving through removal to assure all equipment has been removed or properly secured.

5. Worker safety is paramount during set up and removal operations. The following safety rules will be followed:

- Workers are not allowed to place cones or other devices from the bed of moving pickup trucks.
- Workers must not ride in the bed of a pickup truck during set up or any other time.
- Workers must not place cones or other devices or otherwise work behind a backing work vehicle. 22-8 (Created on 09/15/10)
- If a work vehicle needs to be backing, all workers must be in front of the vehicle and in sight of the driver.
- Violations of these rules will be grounds for disciplinary action, including suspension without pay.

Lighting

1. Temporary lighting should be used in night work zones or when work is at dawn or dusk.
2. Lighting should allow motorists/pedestrians to clearly see work area and potential hazard so they can safely travel through the work area. But make certain that lighting does not "blind" motorists traveling in proximity to the work area.
3. Illumination should allow workers to more clearly see their work activity.
4. Lighting should be directed so motorists can clearly identify locations of workers in the work zone.
5. Overnight procedures to secure work area and protect hazards must be developed and may require auxiliary flashers or lighting that must be activated upon departing the work area.

Overnight Procedures

1. Work locations may require multiple days of activity and may necessitate the securing of the site overnight. The Superintendent and/or foreman will be responsible for making sure that an adequate overnight plan has been established.

2. General Contractor will be consulted if overnight procedures require special equipment, closures, or other contractor involvement.
3. The foreman must determine the proper number of signs and channelizing devices that are needed to alert drivers and pedestrians of the hazards and any changes in the roadway or walkway.
4. All such devices should be reflectorized to ensure good visibility at night. Lighting may be required based upon the hazard and risk.
5. All equipment and supplies should be parked and stored in a safe location where they are not accessible to vehicles or pedestrians.
6. Equipment, vehicles, and materials should be secured to prevent theft and vandalism.
7. Adequate pathways must be established as needed to allow passage to retail or commercial establishments as needed. Hazards should be delineated with cones, barricades, caution tape, snow fence, etc.
8. The foreman is required to conduct a job check to verify that hazards and exposures have been minimized.
9. All caution tape, delineators, cones, barricades, and signs should be checked immediately prior to closing down the job for the day and inspection noted in the daily log by the foreman. Corrective action should be completed before departing the job.

Flagger usage and Training

1. Flaggers are in place to protect themselves, the crew, motorists, and pedestrians traveling through the work zone, and equipment. This is a critical job and one that must be taken seriously.
2. Flaggers must be mentally alert, in good physical condition, be courteous but authoritative.
3. All employees who flag must receive adequate training. Each State has different requirements. Most require some type of training, and many require actual certification. It is the policy that no matter what the State requirements, all employees will be properly trained before flagging.
4. Training and instructions will cover at a minimum the following:
    - Flagger equipment that must be used
    - Layout of the work zone and flagging station
    - Methods to signal traffic to stop, proceed or slow down
    - Methods of one-way control

- Actual demonstration of proper flagging methodology and operations
- Emergency vehicles traveling through the work zone
- Handling emergency situations
- Methods of dealing with hostile drivers
- Flagging procedures when a single flagger is used

5. Written documentation of flagger training will be maintained with other safety training records.
6. Flaggers are to be utilized at locations on a construction site where barricades and warning signs cannot control the moving traffic.
7. Flaggers must wear proper orange, strong yellow-green or yellow warning garments, Class 2 for daytime and Class 3 for nighttime flagging, although the regulations only require Class 2 for nighttime.
8. Proper STOP/SLOW paddles will be used following MUTCD requirements, including reflectorized material when used at night. 18" paddles are the minimum size allowed. 24" paddles should be used for speeds of 45 mph or greater. A flag may be used in emergencies but must be good grade red material at least 24" square and secured to a staff approximately 36" long.
9. Extension staff may be used for paddles for extended periods of flagging. Long staff should be 66" long and accommodate insertion of the 12" staff paddle.
10. Proper and consistent hand and paddle/flag signals must be used as specified in the MUTCD. Flaggers will be tested on appropriate signals and foremen will monitor to assure proper signals are given. Failure to provide proper signals is given. Failure to provide proper signals could expose additional liability in the event of an accident.
11. Flagger stations or positions must be located so that approaching road users have sufficient distance to react and stop at an intended stopping point. Refer to the Stopping Distance Table in the MUTCD.
12. Flagger stations must be illuminated if used at night or reduced light conditions.
13. Advance warning signs, including "flagger" warning signs must be properly positioned for worker safety.
14. Flaggers should stand either in the shoulder adjacent to the road being controlled or in a closed lane prior to stopping road users. The flagger should only stand or move into the lane being used by moving traffic after the traffic has topped.
15. The flagger should stay in line of sight of first road user to assure visibility.

16. For operations requiring more than one flagger, portable radios will be used unless flaggers are close enough to permit voice or hand signals. Hand signals will be verified and agreed at time of initial operation.
17. Flaggers should maintain a steady flow of traffic as much as possible and minimize unnecessary stops. Work vehicles should be held until a safe gap is available whenever possible.
18. Flaggers will be relieved at least every 2 hours for a 15-minute break to assure they remain alert. No worker will be allowed to flag in excess of 10 hours in any day.
19. Drinking water and restroom facilities will be available for flaggers.

Additional Safety Measures

1. Safe access points will be located so workers can get into workspaces without excessive exposure to traffic.

2. If safe parking areas are not available, workers will park at a central location and use work vehicles to reach the work site.

3. Restroom facilities break areas and other facilities will be located to minimize worker exposure to traffic.

4. Entry and exit points for work vehicles and haul trucks will be planned in advance to reduce traffic conflicts and enhance worker safety.

5. Vehicles and equipment with restricted visibility to the rear may not back up in any location where workers or pedestrians are in the area unless suing a spotter.

6. Blind backing (no spotter) will be done only when the driver has determined that the area is free of workers or other vehicles/equipment by physically exiting the vehicle and checking to the rear.

7. Drivers, when departing their vehicles, must be properly attired, including hard hat and appropriate visibility clothing (safety vest).

8. In cold weather, foreman will assure adequate larger sized vests are available for fit over bulky outer clothing.

9. Workers and foremen should never become involved in any altercation with drivers. Simply record the plate number and description of the vehicle and driver so local law enforcement can be notified.

10. Any employee who witnesses a work zone accident must report it immediately to their foreman. This would include incidents involving vehicles, bicyclists, pedestrians, etc.